

Subscription Vending with AVM

Using the official Azure Verified Module `avm/ptn/lz/sub-vending` to create a subscription under a management group and deploy a spoke VNet peered to a central hub — with an Azure DevOps pipeline.

1. What this solution does

This package provisions a complete application landing zone by calling one **Azure Verified Module (AVM)**. AVM modules are Microsoft's official, versioned, tested IaC building blocks published to the public Bicep registry. In a single deployment it:

- **Creates a new subscription** (via a subscription alias) and **places it under a management group**.
- **Creates a resource group** and a **spoke virtual network** with its subnets inside the new subscription.
- **Peers the spoke to a central hub VNet in both directions** — the module builds both sides of the peering for you.
- Optionally **registers resource providers** in the new subscription and waits for it to be ready, removing the readiness race you get when hand-rolling this.

Your `main.bicep` is now just a thin wrapper that forwards values to the module. Everything environment-specific lives in one `.bicepparam` file, so a new landing zone is a single new parameter file.

2. Architecture at a glance

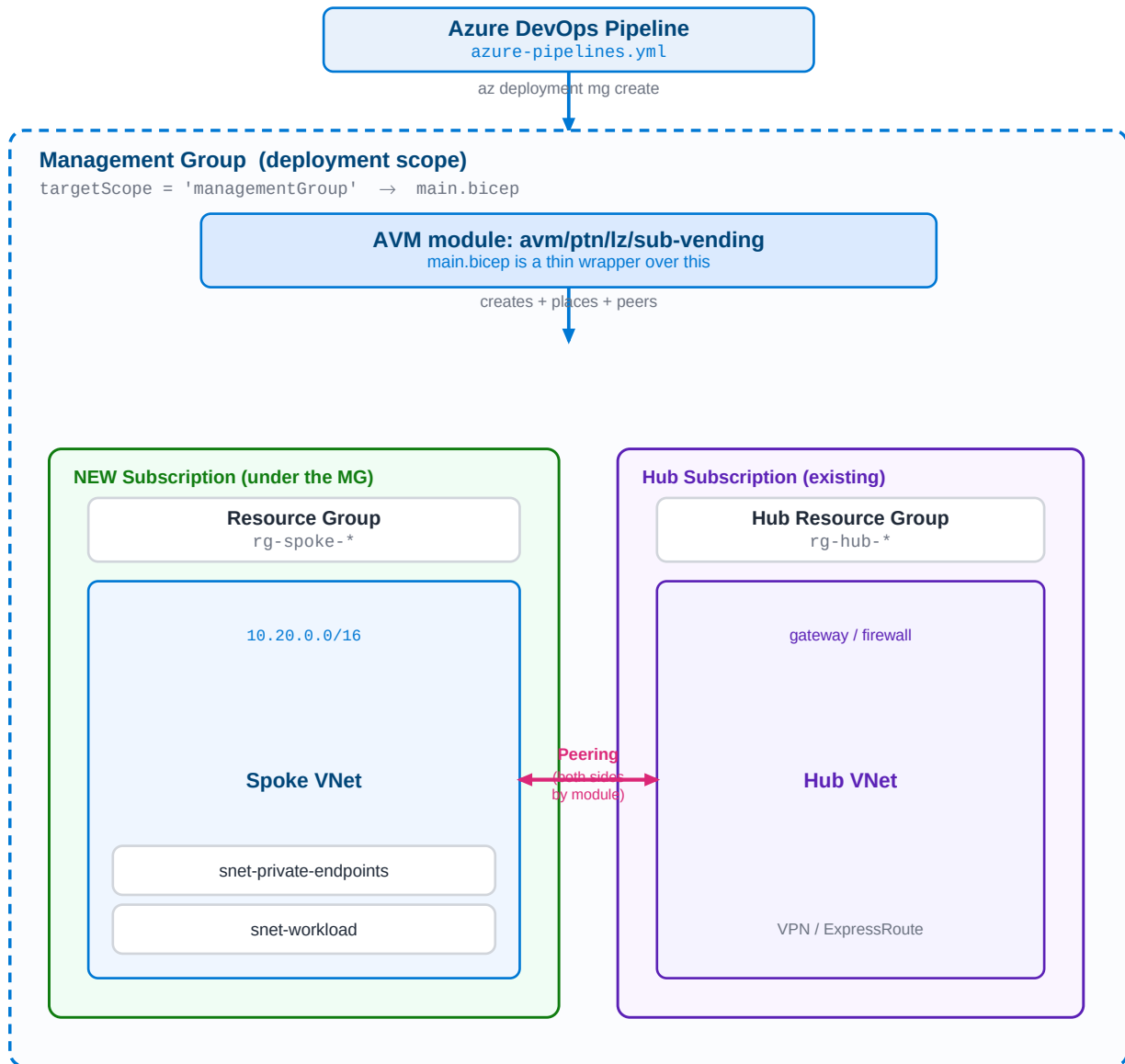


Figure 1 — The pipeline runs a management-group-scoped deployment. `main.bicep` calls the AVM sub-vending module, which creates the subscription under the MG, builds the RG and spoke VNet, and peers it (both directions) to the existing hub subscription.

3. Repository structure

Because the AVM module does the heavy lifting, the previous modules/ folder (vnet, peering, subscription-resources) is gone.

```
repo-root/
|
|-- main.bicep                # MG scope: thin wrapper over the AVM module
|-- bicepconfig.json          # Bicep linter rules (optional)
|
|-- parameters/
|   |-- dev.bicepparam        # <-- copy this to add a new landing zone
|   \-- prod.bicepparam
|
\-- pipelines/
    \-- azure-pipelines.yml    # validate (what-if) + deploy via 'az deployment mg create'
```

4. The code, explained

main.bicep — the wrapper (management group scope)

Declares `targetScope = 'managementGroup'` and calls the AVM module from the public registry, mapping your parameters onto the module's inputs:

```
module subVending 'br/public:avm/ptn/lz/sub-vending:0.3.0' = {
  name: 'lz-${subscriptionAliasName}'
  params: {
    subscriptionAliasEnabled: true
    subscriptionAliasName: subscriptionAliasName
    subscriptionBillingScope: subscriptionBillingScope
    subscriptionManagementGroupAssociationEnabled: true
    subscriptionManagementGroupId: subscriptionManagementGroupId
    virtualNetworkEnabled: true
    virtualNetworkName: virtualNetworkName
    virtualNetworkAddressSpace: virtualNetworkAddressSpace
    virtualNetworkSubnets: virtualNetworkSubnets
    virtualNetworkPeeringEnabled: true
    hubNetworkResourceId: hubNetworkResourceId
    virtualNetworkUseRemoteGateways: useHubGateway
  }
}
```

- `subscriptionManagementGroupAssociationEnabled / ...Id` – create the sub and place it under your MG.
- `virtualNetworkEnabled` + the `virtualNetwork*` params – create the RG, VNet and subnets.
- `virtualNetworkPeeringEnabled` + `hubNetworkResourceId` – build the bidirectional peering to the hub.
- `virtualNetworkUseRemoteGateways` – set true only if the hub has a VPN/ExpressRoute gateway (your `useHubGateway` switch feeds this).

parameters/*.bicepparam — the only file you edit

Targets the template with using ' ../main.bicep' and supplies all values. Note the hub is now a single full **resource ID** (hubNetworkResourceId), not separate name/RG/subscription fields.

pipelines/azure-pipelines.yml

Two stages: **Validate** (az bicep build, which restores the AVM module from the registry, plus a what-if) and **Deploy**. It deploys at management-group scope:

```
az deployment mg create \  
  --name lz-$(Build.BuildId) \  
  --management-group-id $(deploymentManagementGroupId) \  
  --location $(location) \  
  --template-file main.bicep \  
  --parameters parameters/<env>.bicepparam
```

5. Step-by-step deployment

Prerequisites (one-time setup)

#	What	Why
1	ARM service connection in Azure DevOps	Lets the pipeline authenticate to Azure. Note its name.
2	Identity: Owner/Contributor on the deployment management group	Required for the MG-scope deployment and to place the sub under the MG.
3	Identity: a billing role on the billing scope	EA enrollment-account owner, or MCA invoice-section owner. Authorizes subscription creation.
4	Identity: Network Contributor on the hub RG/subscription	The module creates the hub-to-spoke side of the peering there.
5	ADO Environments named 'dev' and 'prod'	Referenced by the deploy job; attach a prod approval gate here.
6	Build agent with outbound internet	Needed so the CLI can restore the AVM module from the public registry.

Pin the module version first

`main.bicep` references `avm/ptn/lz/sub-vending:0.3.0` as a working baseline. The module is updated regularly — **bump this to the latest before deploying**. In VS Code with the Bicep extension, type the `:` after the module path to list versions; or browse the registry tags / the module's GitHub page. If you cross a minor version, re-check the parameter names against that version's README.

Deploy with the pipeline (recommended)

- Push the repo to Azure DevOps.
- Edit `azure-pipelines.yml`: set `azureServiceConnection`, `deploymentManagementGroupId` and `location`.
- Fill in the real values in `parameters/<env>.bicepparam` — especially `subscriptionBillingScope`, `subscriptionManagementGroupId` and `hubNetworkResourceId`.
- In Azure DevOps: **Pipelines** → **New pipeline** → point at `pipelines/azure-pipelines.yml`.
- Run it and choose **dev** or **prod**. Validate runs first; on success, Deploy creates the subscription and landing zone.

Deploy manually from a terminal (alternative)

```

az login

# Preview
az deployment mg what-if \
  --management-group-id mg-landingzones-corp \
  --location eastus \
  --template-file main.bicep \
  --parameters parameters/dev.bicepparam

# Deploy
az deployment mg create \
  --name lz-manual \
  --management-group-id mg-landingzones-corp \
  --location eastus \
  --template-file main.bicep \
  --parameters parameters/dev.bicepparam

```

Add a new landing zone later

- `cp parameters/dev.bicepparam parameters/<name>.bicepparam`
- Change subscription name, MG, VNet names, address space and the hub resource ID.
- Add <name> to the values: list in the pipeline, run, and pick it. No template changes.

6. Things to watch for

Pin the version. Never leave the module on an unintended version. A working baseline is set; bump it deliberately and re-validate after any change.

Non-overlapping address space. The spoke and hub VNet ranges must not overlap or the peering is rejected.

Gateway flag. Only set `useHubGateway = true` when the hub actually has a VPN/ExpressRoute gateway; otherwise peering creation fails.

Hub permissions. Missing Network Contributor on the hub leaves a one-sided (Disconnected) peering. Grant it and re-run.

You get more for free. The same module exposes parameters you aren't using yet — additional VNets, Azure Bastion, NAT gateway, IPAM-based addressing, vWAN topology, role assignments and budgets — without writing more Bicep.